SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA0606 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:100**

Annexure A

Experiments

Code of Conduct:

I SARSHINI .S – 192321113 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SARSHINI.S

Annexure A

1. Design and conduct an experiment to implement a Naïve Bayes classifier on a real-world dataset. Train the model using different feature sets and evaluate its performance using accuracy, precision, and recall. Analyze how assumptions of feature independence affect the results. Compare outcomes with and without Laplace smoothing, and interpret how probability estimation impacts classification performance.

Experiment: Implementation of Naïve Bayes Classifier

Aim

To design and conduct an experiment implementing a Naïve Bayes classifier on a real-world dataset, evaluate its performance using different feature sets, and analyze the impact of feature independence assumptions and Laplace smoothing on classification accuracy.

Dataset Description

A real-world dataset such as the **Iris dataset / Spam Email dataset / Student Performance dataset** is used. The dataset contains multiple features (independent variables) and a target class (dependent variable).

Example:

- Features: Age, Marks, Study Hours, Attendance
- Target: Pass/Fail

Methodology

Step 1: Data Preprocessing

- Handle missing values
- Convert categorical data into numerical format (encoding)
- Normalize or scale features if required
- Split dataset into training and testing sets (e.g., 80:20 ratio)

Step 2: Feature Selection

- Train the model using:
  - Full feature set
  - Reduced feature set (selected important features)
- Compare performance differences

Step 3: Model Training

- Apply Naïve Bayes classifier (Gaussian/Multinomial depending on data type)
- Train the model using training data

Step 4: Prediction

- Use the trained model to predict outputs on test data

Step 5: Performance Evaluation

Evaluate using:

- **Accuracy** = (Correct Predictions / Total Predictions)
- **Precision** = (True Positive / (True Positive + False Positive))
- **Recall** = (True Positive / (True Positive + False Negative))

Theoretical Concept

Naïve Bayes is based on Bayes' Theorem:

It assumes that all features are conditionally independent given the class label.

Experiment Variations

Case 1: Without Laplace Smoothing

- Probability of unseen features may become zero
- Leads to poor classification in some cases

Case 2: With Laplace Smoothing

- Adds a small constant (usually 1) to all probabilities
- Avoids zero probability problem
- Improves robustness of the model

Results and Analysis

1. Effect of Feature Sets

- Full feature set: Higher accuracy but may include redundant features
- Reduced feature set: Faster computation but possible accuracy drop
- If features are highly dependent, Naïve Bayes performance decreases

2. Effect of Feature Independence Assumption

- Naïve Bayes assumes features are independent
- In real-world datasets, features are often correlated
- Violation of independence assumption may reduce accuracy

- However, Naïve Bayes still performs well in many practical cases

3. Effect of Laplace Smoothing

- Without smoothing: Zero probability issues reduce accuracy
- With smoothing: Better generalization and stable predictions

4. Probability Estimation Impact

- Accurate probability estimation improves classification
- Poor estimation leads to misclassification
- Smoothing ensures more realistic probability distribution

Sample Output (Example)

Model Variation	Accuracy	Precision	Recall
Full Features (No Smooth)	85%	83%	80%
Full Features (Smooth)	88%	86%	84%
Reduced Features	82%	80%	78%

Conclusion

The Naïve Bayes classifier is simple, fast, and effective for classification tasks. While the independence assumption is often violated in real-world data, the model still performs reasonably well. Laplace smoothing significantly improves performance by handling zero probabilities. Feature selection also plays a crucial role in optimizing model accuracy and efficiency.

Result

The Naïve Bayes classifier was successfully implemented and evaluated. It was observed that Laplace smoothing and proper feature selection improve classification performance, while violation of feature independence slightly affects accuracy but does not drastically reduce effectiveness.

2. Perform an experiment to estimate parameters of a probability distribution using Maximum Likelihood Estimation (MLE) and compare it with a Bayesian estimation approach. Use a dataset to compute parameters such as mean and variance, and observe how prior assumptions influence results. Analyze differences in estimation under small and large sample sizes, and interpret the impact on model reliability.

Experiment: Parameter Estimation using Maximum Likelihood Estimation (MLE) and Bayesian Estimation

Aim

To estimate the parameters of a probability distribution using Maximum Likelihood Estimation (MLE) and compare it with Bayesian estimation. The experiment analyzes the influence of prior assumptions and evaluates performance under small and large sample sizes.

Dataset Description

A real-world dataset (e.g., **heights of students / exam scores / rainfall data**) is used.

Example dataset:
X = {45, 50, 55, 60, 65, 70, 75}

We assume the data follows a **normal distribution**.

Methodology

Step 1: Data Collection

- Select a dataset representing continuous values
- Divide into:
    - Small sample (first 3–4 values)
    - Large sample (full dataset)

Step 2: MLE Estimation

For a normal distribution:

- Mean ($\hat{\mu}$) = Sample Mean
- Variance ($\hat{\sigma}^2$) = Average of squared deviations

Procedure:

- Compute mean using sample data
- Compute variance using deviations from mean

Observation:

- MLE depends only on observed data
- No prior assumptions are used

Step 3: Bayesian Estimation

- Assume prior distribution for mean (e.g., Normal prior)
- Combine prior with observed data to compute posterior

Procedure:

- Define prior belief (mean and variance)
- Update prior using observed data
- Compute posterior mean and variance

Observation:

- Posterior = Combination of prior + data
- Prior influences estimation, especially with small samples

Experiment Computation

Case 1: Small Sample (n = 3)

Data: {45, 50, 55}

- **MLE Mean** = 50
- **Bayesian Mean** = Depends on prior (e.g., prior mean = 60 → posterior shifts toward 60)

Case 2: Large Sample (n = 7)

Data: {45, 50, 55, 60, 65, 70, 75}

- **MLE Mean** = 60
- **Bayesian Mean** ≈ 60 (less influenced by prior)

Results and Analysis

1. MLE vs Bayesian Estimation

Aspect	MLE	Bayesian Estimation
Basis	Only data	Prior + Data
Flexibility	Less flexible	More flexible
Computation	Simple	More complex
Prior Knowledge	Not used	Used

2. Effect of Prior Assumptions

- Strong prior → heavily influences results
- Weak prior → behaves similar to MLE
- Incorrect prior → may lead to biased estimates

3. Small Sample Analysis

- MLE may be unstable due to limited data
- Bayesian estimation provides better stability
- Prior knowledge improves reliability

4. Large Sample Analysis

- MLE becomes more accurate
- Bayesian estimation converges to MLE
- Prior influence becomes negligible

5. Impact on Model Reliability

- Small dataset:
  - Bayesian method is more reliable
- Large dataset:
  - Both methods perform similarly
- Proper parameter estimation improves prediction accuracy

Conclusion

The experiment demonstrates that MLE is a simple and effective method for parameter estimation when sufficient data is available. Bayesian estimation, however, incorporates prior knowledge and provides more stable estimates, especially for small datasets. As sample size increases, both methods converge, reducing the influence of prior assumptions.

Result

Parameters of the probability distribution were successfully estimated using both MLE and Bayesian approaches. It was observed that Bayesian estimation provides more reliable results for small samples, while MLE performs well for large datasets where prior influence becomes minimal.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**15**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand